

Embedded System:

- An embedded system is an electronic/electro-mechanical system designed to perform a specific function and a combination of both hardware and firmware (software).
- Every embedded system is unique and the hardware as well as the firmware is highly specialized to the application domain.
- Embedded systems are becoming an inevitable part of any product or equipment in all fields including household appliances, telecommunications, medical equipment, industrial control, consumer products, etc.
- Embedded system is a combination of 3 things
 1. Hardware
 2. Software
 3. Mechanical component & it is supposed to do only one specific task only

Examples:**Example1: Washing Machine**

A washing machine from an embedded systems point of view has:

a.Hardware: Buttons, displays & buzzer, electronic circuitry.

b.Software: It has a chip on the circuit that holds the software which drives controls & monitors various operations possible.

c.Mechanical components: the internals of a washing machine which actually wash the clothes control the input and output of water.

Example-2: Air Conditioner

An Air Conditioner from an embedded systems point of view has:

a.Hardware: Remote, display & buzzer, infrared Sensors, electronic circuitry

b.Software: It has a chip on the circuit that holds the software which drives control & monitors the various operations possible. The software monitors the external temperature through the sensors and then releases the coolant or suppresses it.

c. Mechanical components: The internals of an air conditioner the motor, the outlet, etc.-

Differences between General Purpose computing system and Embedded system:

Criteria	General Purpose computing system	Embedded system
Contents	A system which is a combination of a generic hardware and general-purpose operating system for executing a variety of applications	A system which is a combination of a special-purpose hardware and embedded OS for executing a variety of applications
OS	It contains a general-purpose operating system (GPOS)	It may or may not contain an operating system for functioning
Alterations	Applications are alterable (programmable) by the user. (It is possible for end user to re-install the OS and also add or remove user applications)	The firmware of the Embedded system is pre-programmed and it is non-alterable by the end user.
Key Factor	Performance is the key deciding factor in the selection of the system. Faster is better.	Application specific requirements (like performance, power requirements, memory usage etc) are the key deciding factor.
Power consumption	More	Less
Response Time	Not Critical	Critical for some applications
Execution	Need not be deterministic	Deterministic for certain types of ES like 'Hard real time systems'.

Classification of Embedded Systems:

The classification of embedded system is based on following criteria's:

- On generation
- On complexity & performance
- On deterministic behaviour
- On triggering

On generation:

1. First generation (1G):
 - Built around 8bit microprocessor & microcontroller.
 - Simple in hardware circuit & firmware developed.
 - Examples: Digital telephone keypads.
2. Second generation (2G):
 - Built around 16-bit μp & 8-bit μc .
 - They are more complex & powerful than 1G μp & μc .
 - Examples: SCADA systems
3. Third generation (3G):
 - Built around 32-bit μp & 16-bit μc .
 - Concepts like Digital Signal Processors (DSPs), Application Specific Integrated Circuits (ASICs) evolved.
 - Examples: Robotics, Media, etc.
4. Fourth generation:
 - Built around 64-bit μp & 32-bit μc .
 - The concept of System on Chips (SoC), Multicore Processors evolved.
 - Highly complex & very powerful.
 - Examples: Smart Phones.

On complexity & performance:

1. Small-scale:
 - Simple in application need
 - Performance not time-critical.
 - Built around low performance & low cost 8 or 16 bit $\mu p/\mu c$.
 - Example: an electronic toy
2. Medium-scale:
 - Slightly complex in hardware & firmware requirement.
 - Built around medium performance & low cost 16 or 32 bit $\mu p/\mu c$.
 - Usually contain operating system.
 - Examples: Industrial machines.
3. Large-scale:
 - Highly complex hardware & firmware

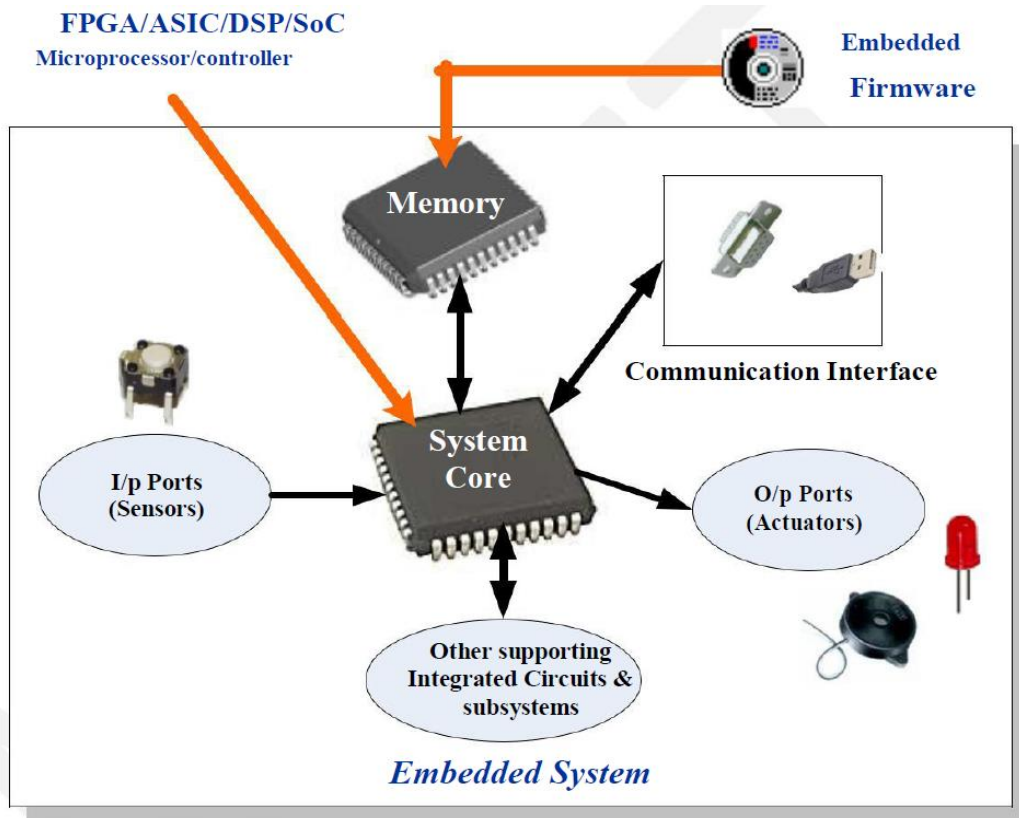
Major Application Areas of ES:

The application areas and the products in the embedded domain are countless. A few of the important domains and products are listed below:

- Consumer electronics: Camcorders, cameras, etc.
- Household appliances: Television, DVD players, washing machine, fridge, microwave oven, etc.
- Home automation and security systems: Air conditioners, sprinklers, intruder detection alarms, closed circuit television cameras, fire alarms, etc.
- Automotive industry: Anti-lock braking systems (ABS), engine control, ignition systems, automatic navigation systems, etc.
- Telecom: Cellular telephones, telephone switches, handset multimedia applications, etc.
- Computer peripherals: Printers, scanners, fax machines, etc.
- Computer networking systems: Network routers, switches, hubs, firewalls, etc.
- Healthcare: Different kinds of scanners, EEG, ECG machines etc.
- Measurement & Instrumentation: Digital multi meters, digital CROs, logic analyzers PLC systems, etc.
- Banking & Retail: Automatic teller machines (ATM) and currency counters, point of sales (POS).
- Card Readers: Barcode, smart card readers, hand held devices, etc.

Elements of an embedded system:

- An embedded system is a combination of 3 things, Hardware Software Mechanical Components and it is supposed to do one specific task only.
- A typical embedded system contains a single chip controller which acts as the master brain of the system. Diagrammatically an embedded system can be represented as follows:
- Embedded systems are basically designed to regulate a physical variable (such as Microwave Oven) or to manipulate the state of some devices by sending some signals to the actuators or devices connected to the output port system (such as temperature in Air Conditioner), in response to the input signal provided by the end users or sensors which are connected to the input ports. Hence the embedded systems can be viewed as a reactive system.



- The control is achieved by processing the information coming from the sensors and user interfaces and controlling some actuators that regulate the physical variable.
- Keyboards, push button, switches, etc. are Examples of common user interface input devices and LEDs, LCDs, Piezoelectric buzzers, etc examples for common user interface output devices for a typical embedded system. The requirement of type of user interface changes from application to application based on domain.
- Some embedded systems do not require any manual intervention for their operation. They automatically sense the input parameters from real world through sensors which are connected at input port. The sensor information is passed to the processor after signal conditioning and digitization. The core of the system performs some predefined operations on input data with the help of embedded firmware in the system and sends some actuating signals to the actuator connect connected to the output port of the system.
- The memory of the system is responsible for holding the code (control algorithm and other important configuration details). There are two types of memories are used in any embedded system. Fixed memory (ROM) is used for storing code or program.

- The user cannot change the firmware in this type of memory. The most common types of memories used in embedded systems for control algorithm storage are OTP, PROM, UVEPROM, EEPROM and FLASH.
- An embedded system without code (i.e. the control algorithm) implemented memory has all the peripherals but is not capable of making decisions depending on the situational as well as real world changes.
- Memory for implementing the code may be present on the processor or may be implemented as a separate chip interfacing the processor
- In a controller based embedded system, the controller may contain internal memory for storing code such controllers are called Micro-controllers with on-chip ROM. eg. Atmel AT89C51.

The Core of the Embedded Systems:

The core of the embedded system falls into any one of the following categories.

- General Purpose and Domain Specific Processors
 - Microprocessors
 - Microcontrollers
 - Digital Signal Processors
- Programmable Logic Devices (PLDs)
- Application Specific Integrated Circuits (ASICs)
- Commercial off the shelf Components (COTS)

General Purpose and Domain Specific Processor:

- Almost 80% of the embedded systems are processor/ controller based.
- The processor may be microprocessor or a microcontroller or digital signal processor, depending on the domain and application.

Microprocessor:

- A silicon chip representing a Central Processing Unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of Instructions, which is specific to the manufacturer
- In general the CPU contains the Arithmetic and Logic Unit (ALU), Control Unit and Working registers
- Microprocessor is a dependant unit and it requires the combination of other hardware like Memory, Timer Unit, and Interrupt Controller etc for proper functioning.

- Intel claims the credit for developing the first Microprocessor unit Intel 4004, a 4 bit processor which was released in Nov 1971. Developers of microprocessors:
- Intel – Intel 4004 – November 1971(4-bit)
- Intel – Intel 4040.
- Intel – Intel 8008 – April 1972.
- Intel – Intel 8080 – April 1974(8-bit).
- Intel – Intel 8085 – 1976.

Microcontroller:

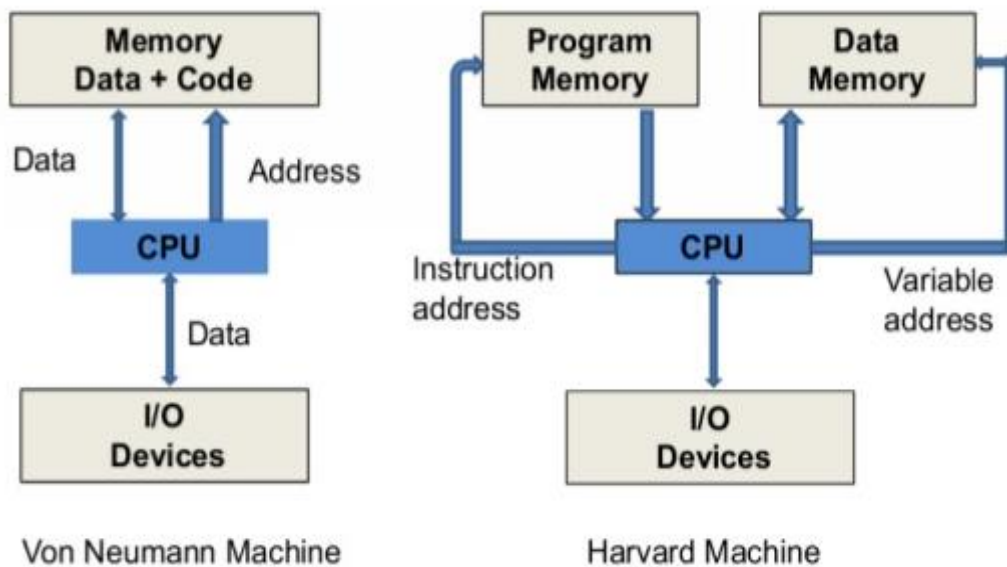
- A highly integrated silicon chip containing a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports
- Microcontrollers can be considered as a super set of Microprocessors
- Microcontroller can be general purpose (like Intel 8051, designed for generic applications and domains) or application specific (Like Automotive AVR from Atmel Corporation. Designed specifically for automotive applications)
- Since a microcontroller contains all the necessary functional blocks for independent working, they found greater place in the embedded domain in place of microprocessors
- Microcontrollers are cheap, cost effective and are readily available in the market
- Texas Instruments TMS 1000 is considered as the world's first microcontroller.

Differences between Microprocessor and Microcontroller:

Microprocessor	Microcontroller
1. Microprocessor is widely used in computer systems.	1. Microcontroller is widely used in embedded systems.
2. It has only a CPU embedded into it	2. It has a CPU, a fixed amount of RAM, ROM and other peripherals all embedded on it.
3. In case of microprocessors we have to connect all the components externally so the circuit becomes large and complex.	3. As all the components are internally connected in microcontroller so the circuit size is less.
4. It consumes more power.	4. It consumes less power than a microprocessor.
5. It has very less internal register storage so it has to rely on external storage. So, all memory based external commands which results in high processing time.	5. It has many registers so processing time is less.

Difference between CISC and RISC Processor:

CISC	RISC
1. Complex instructions.	1. Simple instructions.
2. Main focus is hardware.	2. Main focus is software.
3. Complexity lies in Processor.	3. Complexity lies in Compiler.
4. Multiple clock cycle.	4. Single clock cycle.
5. Transistors are used to store complex instructions.	5. Transistors are used for storing memory.
6. CISC has 100-300 minimum Instructions.	6. RISC uses few instructions (30-40).
7. 8-10 Addressing modes.	7. Few Addressing modes.
8. Variable size/length instructions.	8. Fixed size/length instructions.

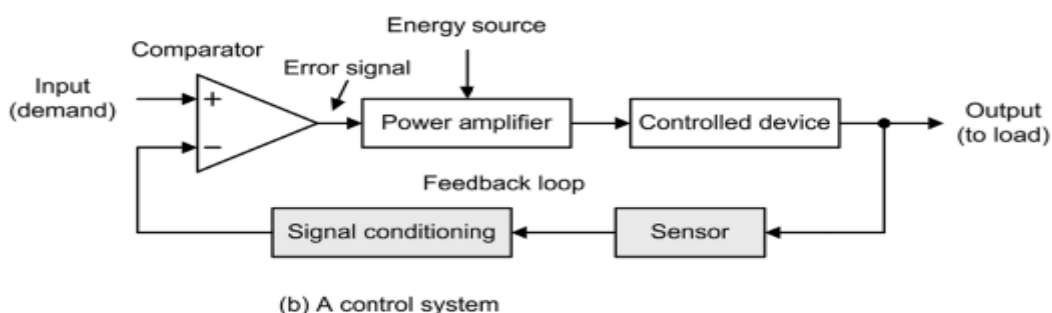
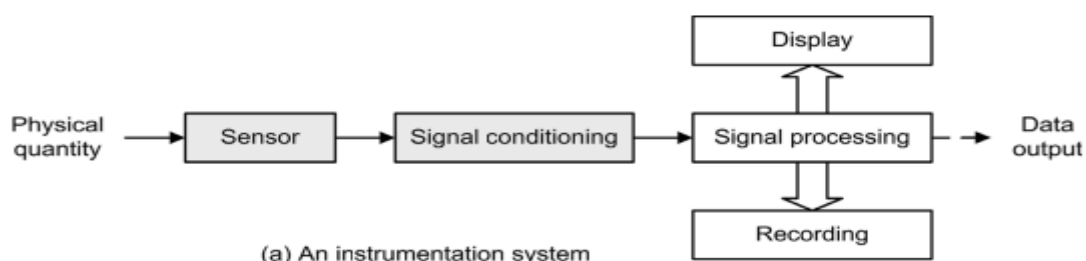
Von Neuman Architecture and Harvard Architecture:

Von-Neuman	Harvard
1. Here the RAM and ROM is not separated, and a single memory connection is given to CPU.	1. Here the CPU is connected separately with RAM and ROM.
2. Less space is required.	2. More space is required.

3. Speed is low because fetching data and instructions at the same time is unable. At a time, it can fetch either a data or instruction.	3. Speed is more because it takes less time to fetch data and instruction at the same time from 2 memories called instruction memory and data memory.
4. There is a common bus for transferring data and instruction.	4. There are separate buses to transfer data and instruction.
5. Used in personal computers and small sized computers.	5. This architecture is used in signal processing and the microcontrollers.
6. Same memory address is used for both instructions and data.	6. Separate physical address is used for both instruction and data.
7. Here the two clock cycles are needed to execute a single instruction.	7. A Single clock is needed to execute a instruction.
8. Instructions and read/write operations cannot be accessed by the CPU at the same time.	8. CPU can access both instructions and write/read operations at the same time.
9. These architecture-based computers are somewhat cheaper than Harvard architecture.	9. This is expensive than Von Neumann architecture.

Instrumentation and control systems:

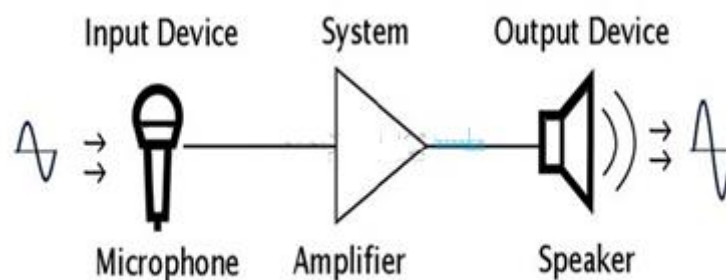
- Instrumentation and control systems Fig.(a) Shows the arrangement of an instrumentation system. The physical quantity to be measured (e.g. temperature) acts upon a sensor that produces an electrical output signal. This signal is an electrical analogue of the physical input but note that there may not be a linear relationship between the physical quantity and its electrical equivalent.



- Fig.(b) shows the arrangement of a control system. This uses negative feedback in order to regulate and stabilize the output. It thus becomes possible to set the input or demand (i.e. what we desire the output to be) and leave the system to regulate itself by comparing it with a signal derived from the output (via a sensor and appropriate signal conditioning).
- A comparator is used to sense the difference in these two signals and where any discrepancy is detected the input to the power amplifier is adjusted accordingly. This signal is referred to as an error signal (it should be zero when the output exactly matches the demand). The input (demand) is often derived from a simple potentiometer connected across a stable d.c. voltage source while the controlled device can take many forms (e.g. a d.c. motor, linear actuator, heater, etc.).

Transducers:

- Transducers are devices that convert energy in the form of sound, light, heat, etc., into an equivalent electrical signal, or vice versa. Before we go further, let's consider a couple of examples that you will already be familiar with. A loudspeaker is a transducer that converts low frequency electric current into audible sounds. A microphone, on the other hand, is a transducer that performs the reverse function i.e. that of converting sound pressure variations into voltage or current. Loudspeakers and microphones can thus be considered as complementary transducers.



- Transducers may be used both as inputs to electronic circuits and outputs from them. From the two previous examples, it should be obvious that a loudspeaker is an output transducer designed for use in conjunction with an audio system. A microphone is an input transducer designed for use with a recording or sound reinforcing system.

Sensors:

- A sensor is a special kind of transducer that converts energy from one form to another for any measurement or control purpose ex. A Temperature sensor. The signal

produced by a sensor is an electrical analogy of a physical quantity, such as distance, velocity, acceleration, temperature, pressure, light level, etc. The signals returned from a sensor, together with control inputs from the user or controller (as appropriate) will subsequently be used to determine the output from the system. The choice of sensor is governed by a number of factors including accuracy, resolution, cost and physical size.

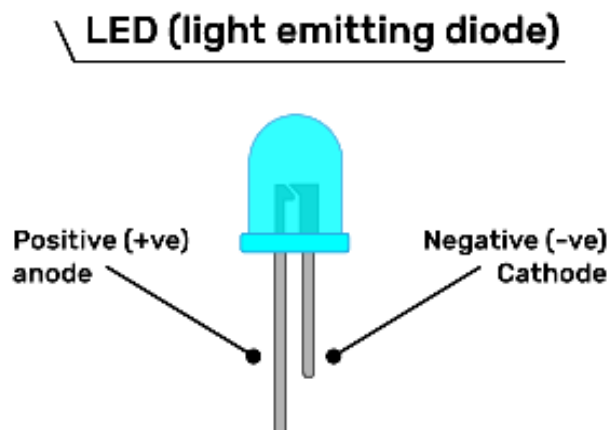
- Sensors can be categorized as either active or passive. An active sensor generates a current or voltage output. A passive transducer requires a source of current or voltage and it modifies this in some way (e.g. by virtue of a change in the sensor's resistance). The result may still be a voltage or current but it is not generated by the sensor on its own.
- Sensors can also be classed as either digital or analogue. The output of a digital sensor can exist in only two discrete states, either 'on' or 'off', 'low' or 'high', 'logic 1' or 'logic 0', etc. The output of an analogue sensor can take any one of an infinite number of voltage or current levels. It is thus said to be continuously variable. Table 15.3 provides details of some common types of sensor.

Actuator:

Actuator is used for output. It is a transducer that may be either mechanical or electrical which converts signals to corresponding physical actions.

LED (Light Emitting Diode):

LED is a p-n junction diode and contains a CATHODE and ANODE for functioning the anode is connected to +ve end of power supply and cathode is connected to -ve end of power supply. The maximum current flowing through the LED is limited by connecting a resistor in series between the power supply and LED as shown in the figure below

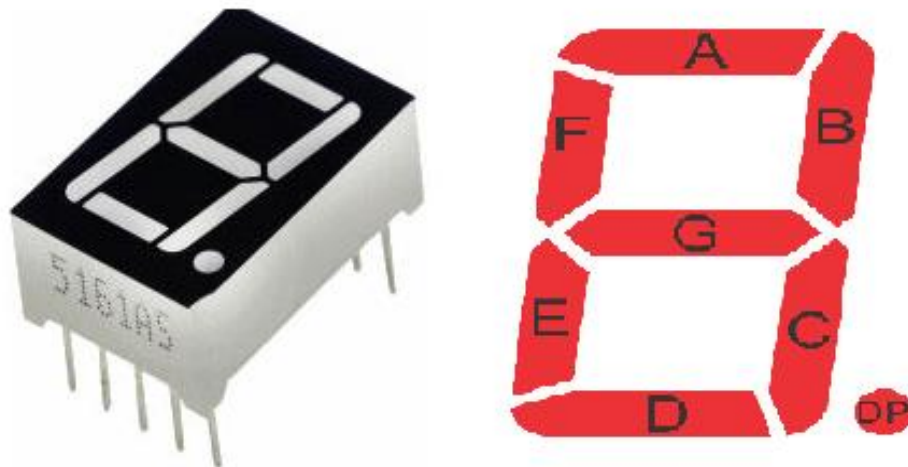


There are two ways to interface an LED to a microprocessor/microcontroller:

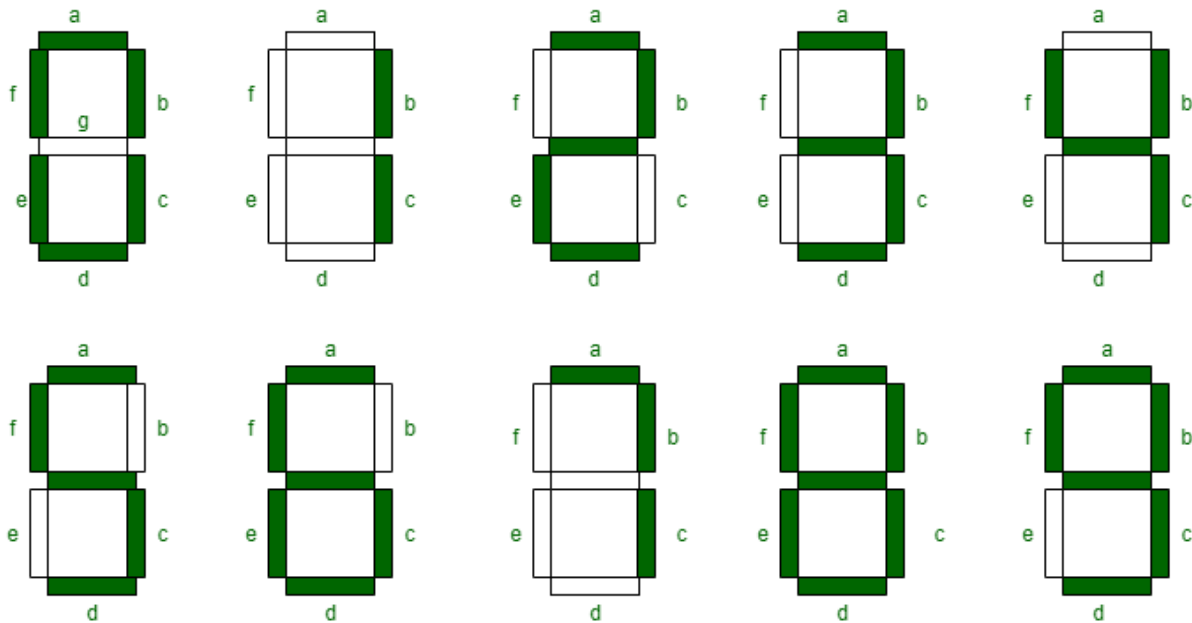
- The Anode of LED is connected to the port pin and cathode to Ground: In this approach the port pin sources the current to the LED when it is at logic high (ie. 1).
- The Cathode of LED is connected to the port pin and Anode to Vcc : In this approach the port pin sources the current to the LED when it is at logic high (ie. 1). Here the port pin sinks the current and the LED is turned ON when the port pin is at Logic low (ie. 0)

7-segment display:

A seven-segment display (SSD), or seven-segment indicator, is a form of electronic display device for displaying decimal numerals that is an alternative to the more complex dot matrix displays. Seven-segment displays are widely used in digital clocks, electronic meters, basic calculators, and other electronic devices that display numerical information.



- The seven elements of the display can be lit in different combinations to represent the Arabic numerals. Often the seven segments are arranged in an oblique (slanted) arrangement, which aids readability.
- In most applications, the seven segments are of nearly uniform shape and size (usually elongated hexagons, though trapezoids and rectangles can also be used), though in the case of adding machines, the vertical segments are longer and more oddly shaped at the ends in an effort to further enhance readability.
- The segments of a 7-segment display are referred to by the letters A to G, where the optional decimal point (an "eighth segment", referred to as DP) is used for the display of non-integer numbers.

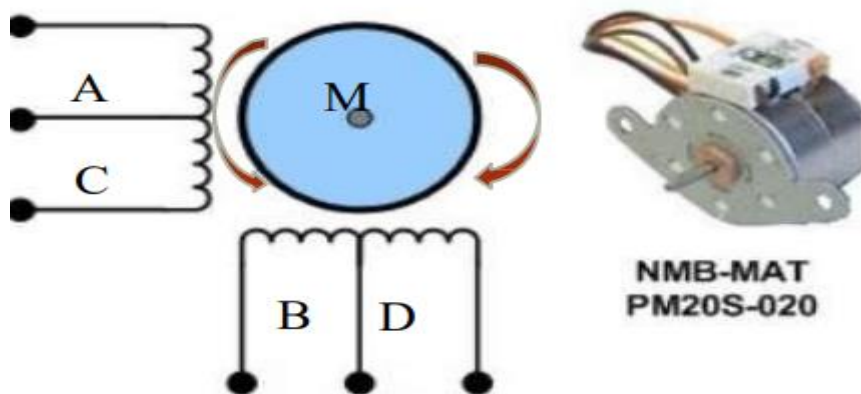


Stepper motor:

- A stepper motor is an electro-mechanical device which generates discrete displacement (motion) in response to dc electrical signals.
- It differs from the normal dc motor in its operation.
- The dc motor produces continuous rotation on applying dc voltage whereas a stepper motor produces discrete rotation in response to the dc voltage applied to it.
- Stepper motors are widely used in industrial embedded applications, consumer electronic products and robotics control systems.
- The paper feed mechanism of a printer/fax makes use of stepper motors for its functioning.
- Based on the coil winding arrangements, a two-phase stepper motors classified into two. They are:
 - Unipolar
 - Bipolar
- A unipolar stepper motor contains two windings per phase.

The direction of rotation (clockwise or anticlockwise) of a stepper motor is controlled by changing the direction of current flow.

Unipolar Stepper Motor

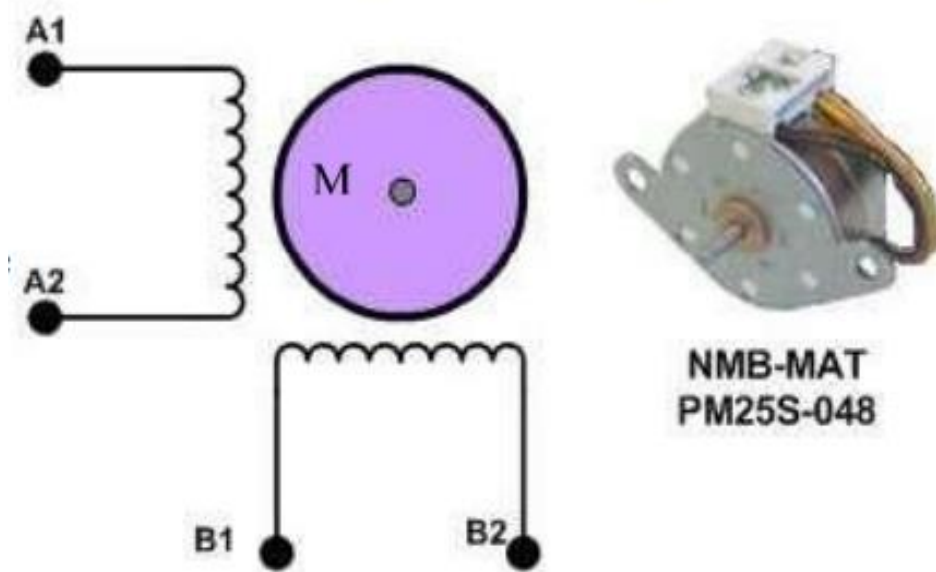


- Current in one direction flows through one coil and in the opposite direction flows through the other coil.
- It is easy to shift the direction of rotation by just switching the terminals to which the coils are connected.
- Figure illustrates the working of a two-phase unipolar stepper motor

Bipolar:

- A bipolar stepper motor contains single winding per phase.
- For reversing the motor rotation the current flow through the windings is reversed dynamically.
- It requires complex circuitry for current flow reversal.

Bipolar Stepper Motor



- There is one disadvantage of unipolar motors. The torque generated by them is quite less. This is because the current is flowing only through the half the winding. Hence they are used in low torque applications.
- On the other hand, bipolar stepper motors are a little complex to wire as we have to use a current reversing H bridge driver IC like an L293D. But the advantage is that the current will flow through the full coil. The resulting torque generated by the motor is larger as compared to a unipolar motor.
- The stepping of stepper motor can be implemented in different ways by changing the sequence of activation of the stator windings. The different stepping modes supported by stepper motor are explained below.
- **Full Step:** In the step mode both the phases are energized simultaneously. The coils A, B, C and D are energized in the following order:
- It should be noted that out of the two windings, only one winding of a phase is energized at a time

Step	Coil A	Coil B	Coil C	Coil D
1	H	H	L	L
2	L	H	H	L
3	L	L	H	H
4	H	L	L	H

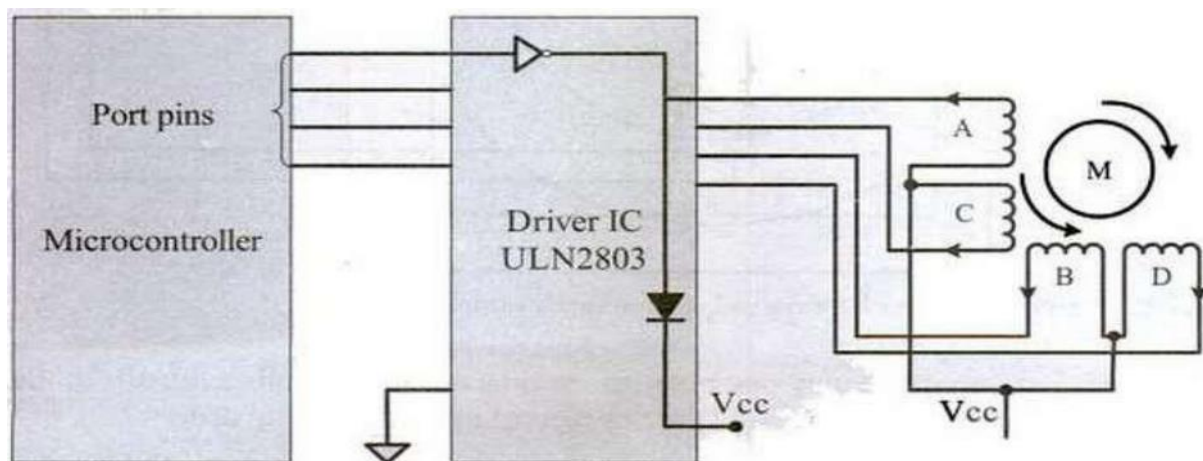
- **Wave Step:** In the wave step mode only one phase is energized at a time and each coils of the phase is energies alternatively. The coils A, B, C and D are energized in the following order:

Step	Coil A	Coil B	Coil C	Coil D
1	H	L	L	L
2	L	H	L	L
3	L	L	H	L
4	L	L	L	H

- **Half Step:** It uses the combination of wave and full step. It has the highest torque and stability. The coil energizing sequence for half step is given below.

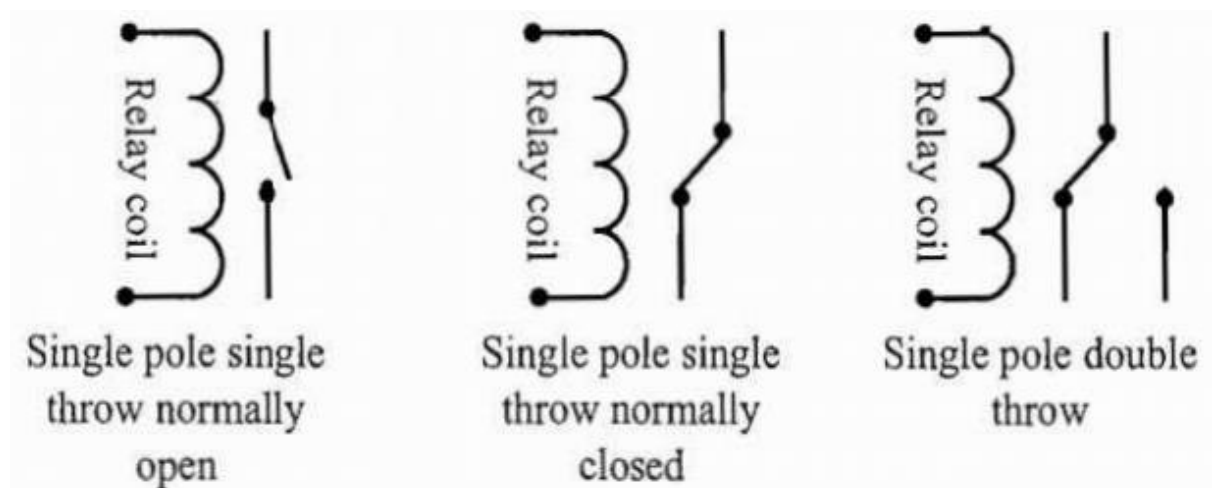
Step	Coil A	Coil B	Coil C	Coil D
1	H	L	L	L
2	H	H	L	L
3	L	H	L	L
4	L	H	H	L
5	L	L	H	L
6	L	L	H	H
7	L	L	L	H
8	H	L	L	H

- The rotation of the stepper motor can be reversed by reversing the order in which the coil is energized. Two-phase unipolar stepper motors are the popular choice for embedded applications.
- The current requirement for stepper motor is little high and hence the port pins of a microcontroller/processor may not be able to drive them directly.
- Also the supply voltage required to operate stepper motor varies normally in the range 5V to 24 V.
- Depending on the current and voltage requirements, special driving circuits are required to interface the stepper motor with microcontroller/processors.
- The following circuit diagram illustrates the interfacing of a stepper motor through a driver circuit connected to the port pins of a microcontroller/processor.



Relay:

- Relay is an electro-mechanical device. In embedded application, the 'Relay' unit acts as dynamic path selectors for signals and power.
- The 'Relay' unit contains a relay coil made up of insulated wire on a metal core and a metal armature with one or more contacts.
- 'Relay' works on electromagnetic principle. When a voltage is applied to the relay coil, current flows through the coil, which in turn generates a magnetic field.
- The magnetic field attracts the armature core and moves the contact point. The movement of the contact point changes the power/signal flow path.
- 'Relays' are available in different configurations. Figure given below illustrates the widely used relay configurations for embedded applications.



- The Single Pole Single Throw configuration has only one path for information flow. The path is either open or closed in normal condition.
- For normally Open Single Pole Single Throw relay, the circuit is normally open and it becomes closed when the relay is energized.
- For normally closed Single Pole Single Throw configuration, the circuit is normally closed and it becomes open when the relay is energized. For Single Pole Double Throw Relay, there are two paths for information flow and they are selected by energizing or de-energizing the relay.

Piezo Buzzer:

- Piezo buzzer is a piezoelectric device for generating audio indications in embedded application.

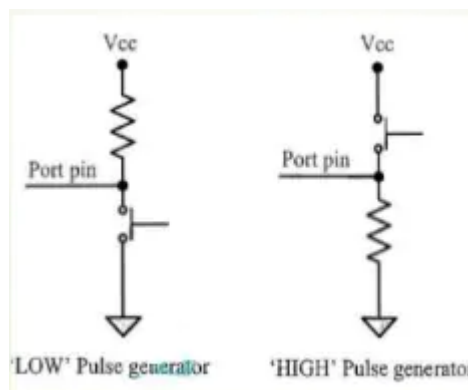
- A piezoelectric buzzer contains a piezoelectric diaphragm which produces audible sound in response to the voltage applied to it.
- Piezoelectric buzzers are available in two types: 'Self-driving' and 'External driving'.
- The 'Self-driving' circuit contains all the necessary components to generate sound at a predefined tone. It will generate a tone on applying the voltage.



- External driving piezo buzzers supports the generation of different tones. The tone can be varied by applying a variable pulse train to the piezoelectric buzzer.

Push Button Switch:

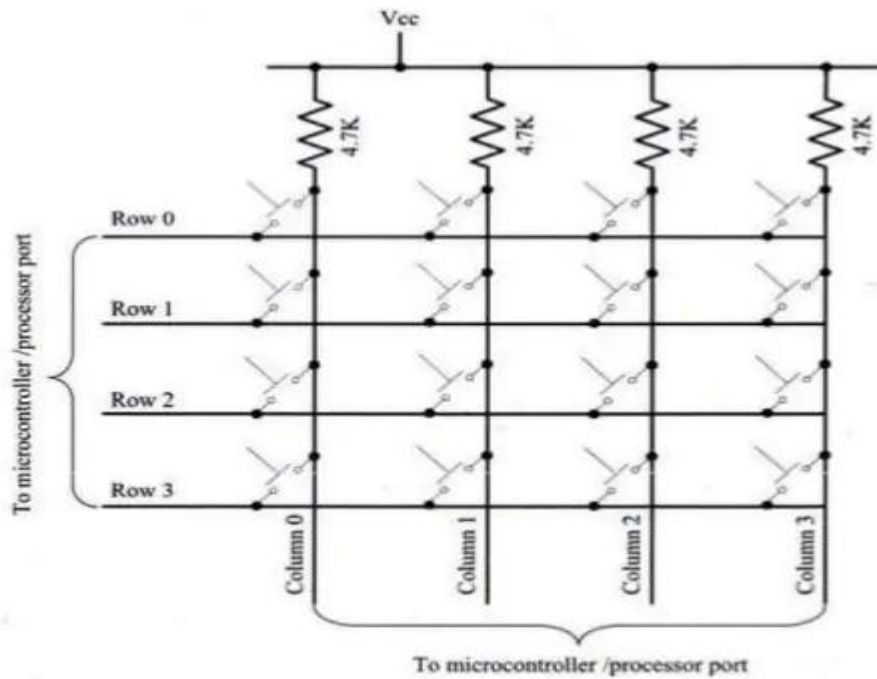
- It is an input device. Push button switch comes in two configurations, namely 'Push to Make' and 'Push to Break'.



- In the 'Push to Make' configuration, the switch is normally in the open state and it makes a circuit contact when it is pushed or pressed.
- In the 'Push to Break' configuration, the switch is normally in the closed state and it breaks the circuit contact when it is pushed or pressed.
- In the embedded application push button is generally used as reset and start switch.

Keyboard:

- Keyboard is an input device for user interfacing. If the number of keys required is very limited, push button switches can be used and they can be directly interfaced to the port pins for reading.



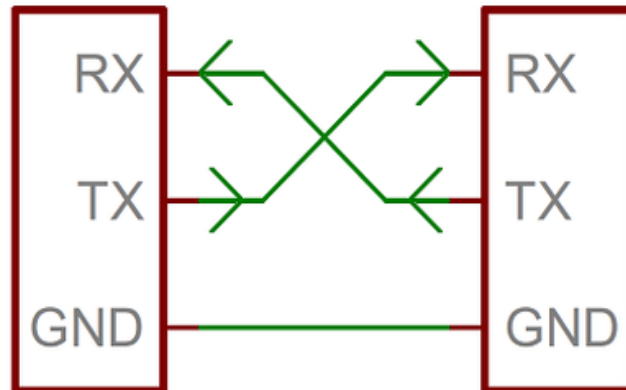
- Matrix keyboard is an optimum solution for handling large key requirements. It greatly reduces the number of interface connections.
- For example, for interfacing 16 keys, in the direct interfacing technique 16 port pins are required, whereas the matrix keyboard only 8 lines are required. The 16 keys are arranged in a 4 column*4 row matrix.
- In a matrix keyboard the keys are arranged in matrix fashion. For detecting a key press, the keyboard uses the scanning technique, where each row of the matrix is pulled low and the columns are read.
- After reading the status of each column corresponding to a row. The row is pulled high and the next row is pulled low and the status of the columns is read.
- This process is repeated until the scanning for all rows is completed. When a row is pulled low and if a key connected to the row is pressed, reading the column to which the key is connected will give logic 0.

Communication Interface:

UART:

- In UART communication, two UARTs communicate directly with each other. The transmitting UART converts parallel data from a controlling device like a CPU into serial form, transmits it in serial to the receiving UART, which then converts the serial data back into parallel data for the receiving device.

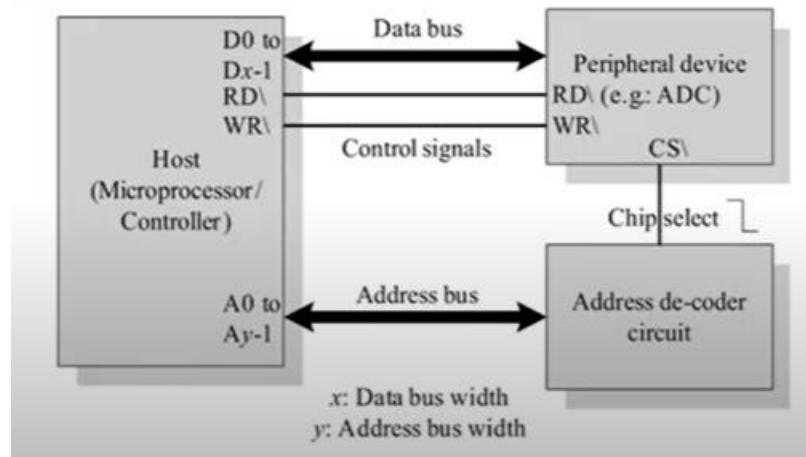
- Only two wires are needed to transmit data between two UARTs. Data flows from the Tx pin of the transmitting UART to the Rx pin of the receiving UART:



- UARTs transmit data asynchronously, which means there is no clock signal to synchronize the output of bits from the transmitting UART to the sampling of bits by the receiving UART.
- Instead of a clock signal, the transmitting UART adds start and stop bits to the data packet being transferred. These bits define the beginning and end of the data packet so the receiving UART knows when to start reading the bits.
- When the receiving UART detects a start bit, it starts to read the incoming bits at a specific frequency known as the baud rate. Baud rate is a measure of the speed of data transfer, expressed in bits per second (bps). Both UARTs must operate at about the same baud rate. The baud rate between the transmitting and receiving UARTs can only differ by about 10% before the timing of bits gets too far off.

Parallel Interface:

- The on board parallel interface is normally used for or communicating with parallel devices which are memory mapped to the host of the system.
- The host processor /controller of the embedded system contain a parallel bus and the device which supports parallel bus can directly connect to this bus system.
- The communication through the parallel bus is controlled by the control signal interface between the device and the host.
- The control signals for communication includes read /write signal and device select signal. The device normally contains device select line and the device becomes active only when this line is asserted by the host processor.



- The direction of data transfer can be controlled through the control signal lines for Read and Write. Only the host processor has control over the Read and Write control signal. The device is normally memory mapped to the host processor and range of address is assigned to it.
- An address Decoder circuit is used for generating the chip select signal for the device. When the address selected by the processor is within the range assigned for the device, the Decoder circuits activates the chip select line and thereby the device becomes active. The processor then can read or write from or to the device by asserting the corresponding control line

USB:

- USB, short for Universal Serial Bus, is a standard type of connection for many different kinds of devices. Generally, USB refers to the types of cables and connectors used to connect these many types of external devices to computers.



- The Universal Serial Bus standard has been extremely successful. USB ports and cables are used to connect hardware such as printers, scanners, keyboards, mice, flash

drives, external hard drives, joysticks, cameras, and more to computers of all kinds, including desktops, tablets, laptops, net books, etc.

- In fact, USB has become so common that you'll find the connection available on nearly any computer-like device such as video game consoles, home audio/visual equipment, and even in many automobiles.
- Many portable devices, like smart phones, eBook readers, and small tablets, use USB primarily for charging. USB charging has become so common that it's now easy to find replacement electrical outlets at home improvement stores with USB ports built in, negating the need for a USB power adapter.

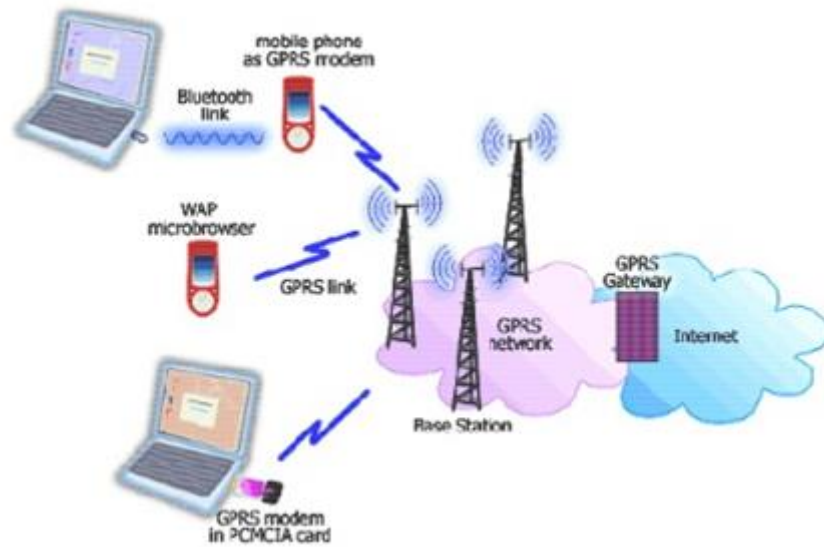
Wi-Fi:

- Wi-Fi is a technology for wireless local area networking with devices based on the IEEE 802.11 standards. Wi-Fi is a trademark of the Wi-Fi Alliance, which restricts the use of the term Wi-Fi Certified to products that successfully complete interoperability certification testing.
- Devices that can use Wi-Fi technology include personal computers, video-game consoles, phones and tablets, digital cameras, smart TVs, digital audio players and modern printers.
- Wi-Fi compatible devices can connect to the Internet via a WLAN and a wireless access point. Such an access point (or hotspot) has a range of about 20 meters (66 feet) indoors and a greater range outdoors. Hotspot coverage can be as small as a single room with walls that block radio waves, or as large as many square kilometres achieved by using multiple overlapping access points.

General Packet Radio Service:

- GPRS is a packet oriented mobile data service on the 2G and 3G cellular communication system's global system for mobile communications (GSM). GPRS was originally standardized by European Telecommunications Standards Institute (ETSI) in response to the earlier CDPD and i-mode packet-switched cellular technologies. It is now maintained by the 3rd Generation Partnership Project (3GPP).
- GPRS usage is typically charged based on volume of data transferred, contrasting with circuit switched data, which is usually billed per minute of connection time.

Sometimes billing time is broken down to every third of a minute. Usage above the bundle cap is charged per megabyte, speed limited, or disallowed.



- GPRS is a best-effort service, implying variable throughput and latency that depend on the number of other users sharing the service concurrently, as opposed to circuit switching, where a certain quality of service (QoS) is guaranteed during the connection. In 2G systems, GPRS provides data rates of 56–114 kbit/second. 2G cellular technology combined with GPRS is sometimes described as 2.5G, that is, a technology between the second (2G) and third (3G) generations of mobile telephony. It provides moderate-speed data transfer, by using unused time division multiple access (TDMA) channels in, for example, the GSM system. GPRS is integrated into GSM Release 97 and newer releases.